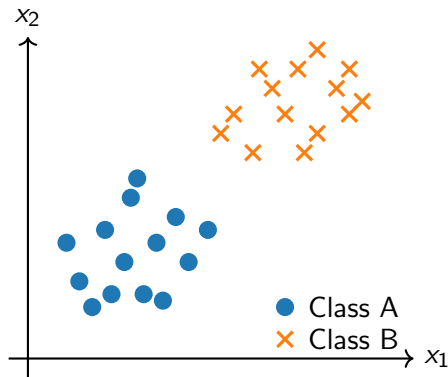


From Gini Impurity to Random Forests

Decision Trees and Ensemble Methods

Binary Classification Setup



Goal: Learn a function
 $f : \mathbb{R}^d \rightarrow \{A, B\}$

Given training data with features and labels, find decision boundaries that separate the classes.

Key question: How do we decide where to split?

Gini Impurity: Measuring “Mixedness”

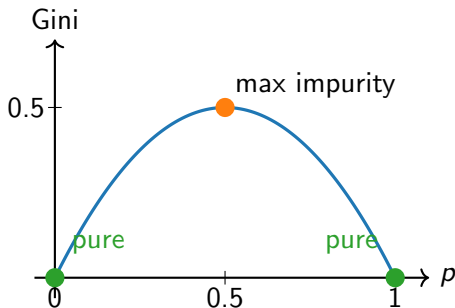
Definition:

$$\text{Gini}(S) = 1 - \sum_{i=1}^k p_i^2$$

For binary classification (p = proportion of class A):

$$\text{Gini}(S) = 1 - p^2 - (1 - p)^2$$

- Gini = 0: Pure node (all one class)
- Gini = 0.5: Maximum impurity (50-50 split)



Gini impurity is maximized when classes are equally mixed

Worked Example: Dataset

Sample	Feature A	Feature B	Label
1	0	1	0
2	0	0	0
3	0	1	0
4	1	0	0
5	1	1	1
6	1	0	1
7	1	1	1
8	1	1	1

Label distribution: 4 samples with Label=0, 4 samples with Label=1

Goal: Find the best feature to split on at the root node

Parent Node Gini Impurity

Class proportions:

$$p_0 = \frac{4}{8} = 0.5, \quad p_1 = \frac{4}{8} = 0.5$$

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Gini calculation:

$$\text{Gini}(\text{parent}) = 1 - 0.5^2 - 0.5^2 = 1 - 0.25 - 0.25 = 0.5$$

Parent Node Gini Impurity

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Gini calculation:

$$\text{Gini}(\text{parent}) = 1 - 0.5^2 - 0.5^2 = 1 - 0.25 - 0.25 = 0.5$$

Interpretation: Maximum impurity (50/50 split between classes)

Split on Feature A

Left child (Feature A = 0): Samples 1, 2, 3 $\rightarrow$ Labels: 0, 0, 0

$$\text{Gini}(\text{left}) = 1 - 1^2 - 0^2 = 0 \quad (\text{pure node})$$

Split on Feature A

Left child (Feature A = 0): Samples 1, 2, 3 $\rightarrow$ Labels: 0, 0, 0

$$\text{Gini}(\text{left}) = 1 - 1^2 - 0^2 = 0 \quad (\text{pure node})$$

Right child (Feature A = 1): Samples 4, 5, 6, 7, 8 $\rightarrow$ Labels: 0, 1, 1, 1, 1

$$p_0 = 0.2, \quad p_1 = 0.8$$

$$\text{Gini}(\text{right}) = 1 - 0.2^2 - 0.8^2 = 0.32$$

Split on Feature A

Left child (Feature A = 0): Samples 1, 2, 3 $\rightarrow$ Labels: 0, 0, 0

$$\text{Gini}(\text{left}) = 1 - 1^2 - 0^2 = 0 \quad (\text{pure node})$$

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$$p_0 = 0.2, \quad p_1 = 0.8$$

$$\text{Gini}(\text{right}) = 1 - 0.2^2 - 0.8^2 = 0.32$$

Weighted Gini:

$$\text{Gini}_A = \frac{3}{8} \times 0 + \frac{5}{8} \times 0.32 = 0.2$$

Split on Feature B

Left child (Feature B = 0): Samples 2, 4, 6 $\rightarrow$ Labels: 0, 0, 1

$$\text{Gini}(\text{left}) = 1 - \left(\frac{2}{3}\right)^2 - \left(\frac{1}{3}\right)^2 = \frac{4}{9} \approx 0.444$$

Split on Feature B

Left child (Feature B = 0): Samples 2, 4, 6 $\rightarrow$ Labels: 0, 0, 1

$$\text{Gini(left)} = 1 - \left(\frac{2}{3}\right)^2 - \left(\frac{1}{3}\right)^2 = \frac{4}{9} \approx 0.444$$

Right child (Feature B = 1): Samples 1, 3, 5, 7, 8 $\rightarrow$ Labels: 0, 0, 1, 1, 1

$$\text{Gini(right)} = 1 - 0.4^2 - 0.6^2 = 0.48$$

Split on Feature B

Left child (Feature B = 0): Samples 2, 4, 6 $\rightarrow$ Labels: 0, 0, 1

$$\text{Gini}(\text{left}) = 1 - \left(\frac{2}{3}\right)^2 - \left(\frac{1}{3}\right)^2 = \frac{4}{9} \approx 0.444$$

Right child (Feature B = 1): Samples 1, 3, 5, 7, 8 $\rightarrow$ Labels: 0, 0, 1, 1, 1

$$\text{Gini}(\text{right}) = 1 - 0.4^2 - 0.6^2 = 0.48$$

Weighted Gini:

$$\text{Gini}_B = \frac{3}{8} \times 0.444 + \frac{5}{8} \times 0.48 = 0.467$$

Comparison & Decision

Split	Weighted Gini
Feature A	0.200
Feature B	0.467

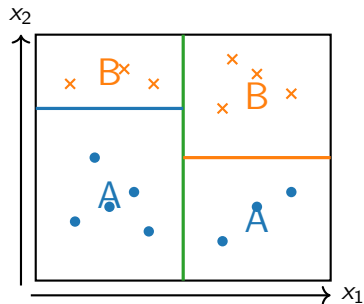
Comparison & Decision

Split	Weighted Gini
Feature A	0.200
Feature B	0.467

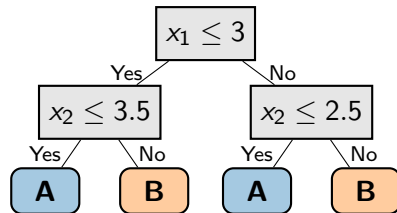
Decision: Split on Feature A (lower Gini impurity)

Feature A perfectly separates all Label=0 samples, while Feature B mixes classes in both children.

Building the Tree



Feature space partitioning



Corresponding decision tree

Each split uses Gini to find the best feature and threshold

Bootstrap Sampling: Creating Diverse Training Sets

Original Data (8 samples)



↓ sample with replacement

Sample 1: 1 1 3 5 5 6 7 8 (1,5 appears twice, 2,4 missing)

Sample 2: 2 2 3 4 4 6 7 8 (2,4 appear twice, 1,5 missing)

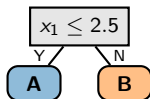
Sample 3: 1 3 3 5 6 7 8 8 (3,8 appear twice, 2,4 missing)

Each tree sees **different data** → learns **different patterns** → makes **different errors**

Different Data → Different Trees

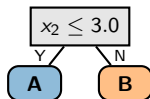
Bootstrap Sample 1

More A samples



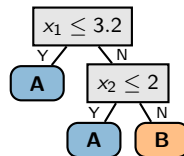
Bootstrap Sample 2

More B samples



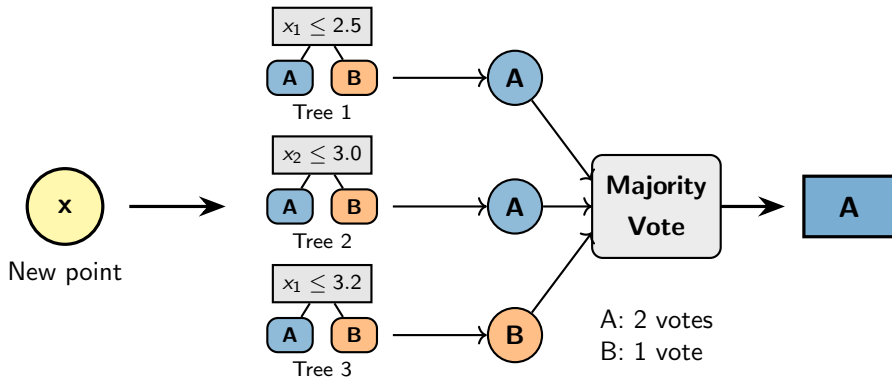
Bootstrap Sample 3

Different mix



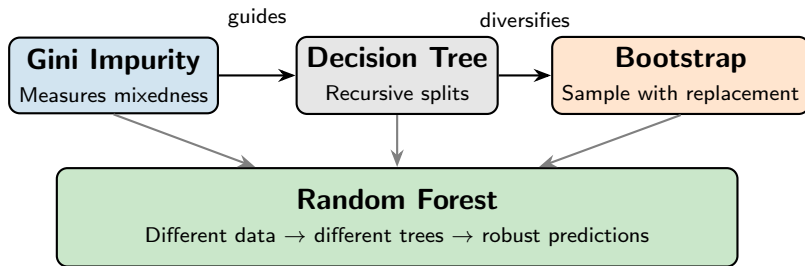
Same algorithm + different data → **different splits and tree structures**

Aggregation: Majority Voting



Individual trees may err, but the **ensemble** is robust:
uncorrelated errors cancel out through voting

Summary



Key Takeaways:

1. Gini impurity ($1 - \sum p_i^2$) quantifies class mixing; lower = purer
2. Trees greedily select splits that minimize weighted Gini
3. Bootstrap sampling creates diverse training sets for each tree
4. Aggregating diverse trees reduces variance without increasing bias

Core principle: Diversity through bootstrapping transforms high-variance trees into a low-variance ensemble.